Lecture_01_Introduction_to_Computer_Graphics

Lecture 1 — Introduction to Computer Graphics

Lecture focus: What computer graphics is, why it matters, where it is used, and how this course is organized.

Learning outcomes

1. Define computer graphics in a precise but practical way.
2. Distinguish between raster images and vector graphics.
3. Explain the difference between modeling and rendering.
4. Identify major application areas of computer graphics.
5. Describe the basic idea of the graphics pipeline.
6. Recognize how computer graphics connects with mathematics, algorithms, hardware, and human perception.

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1. What is computer graphics?

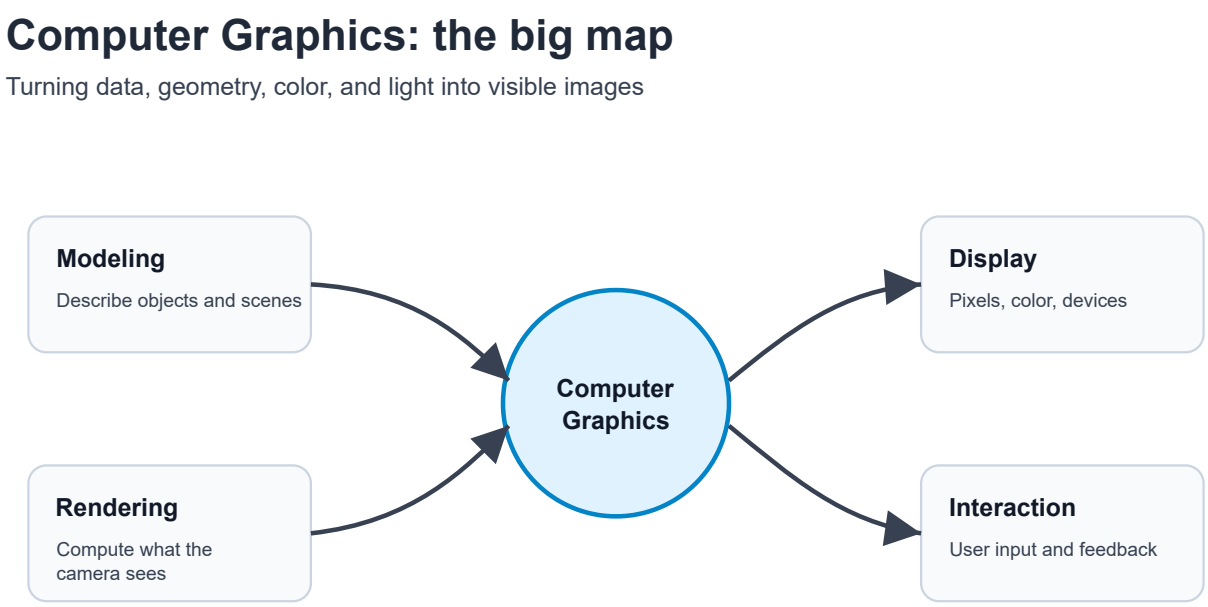
Computer graphics is the field of computer science concerned with **creating, storing, manipulating, and displaying visual content using computers**.

A simple definition is:

> **Computer graphics is the use of computation to generate or control images.**

The word *image* here should be interpreted broadly. It may mean:

- a still picture,
- a graph or chart,
- a line drawing,
- a user interface,
- a 2D animation,
- a 3D game scene,
- a scientific visualization,
- a medical scan rendering,
- a virtual reality environment,
- or a full movie frame generated partly or completely by computer.



Computer graphics often begins with **abstract descriptions**: points, lines, curves, polygons, surfaces, colors, materials, lights, cameras, and transformations. These descriptions are processed by algorithms and eventually converted into visible pixels on a display.

So computer graphics is not only about drawing. It is about answering questions such as:

- How do we represent a shape mathematically?
- How do we draw a line using only pixels?
- How do we rotate an object on screen?
- How do we show a 3D world on a 2D screen?
- How do we decide which object is visible and which object is hidden?
- How do we create the illusion of motion?
- How do we make synthetic objects look realistic, stylized, or informative?

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2. Why study computer graphics?

Computer graphics is one of the most visible branches of computer science. Students experience its results every day: phone interfaces, maps, games, videos, websites, animations, digital art, simulations, and scientific plots.

However, behind these visual outputs there are deep computational ideas:

- **Geometry:** points, vectors, matrices, curves, surfaces.
- **Algorithms:** line drawing, clipping, filling, hidden-surface removal.

- **Linear algebra:** transformations, projections, coordinate systems.
- **Data structures:** meshes, scene graphs, spatial indexing.
- **Hardware:** displays, GPUs, frame buffers, graphics memory.
- **Physics:** light, reflection, shading, motion.
- **Human perception:** color, depth, contrast, motion, visual attention.

That combination makes computer graphics a highly interdisciplinary subject.

Small classroom question

Suppose you see a rotating 3D cube on a 2D monitor. Is the cube really 3D?

The answer is: **the object is represented as 3D data, but the final display is a 2D image.** The computer creates the illusion of 3D by projecting the 3D model onto a 2D screen and updating the image quickly over time.

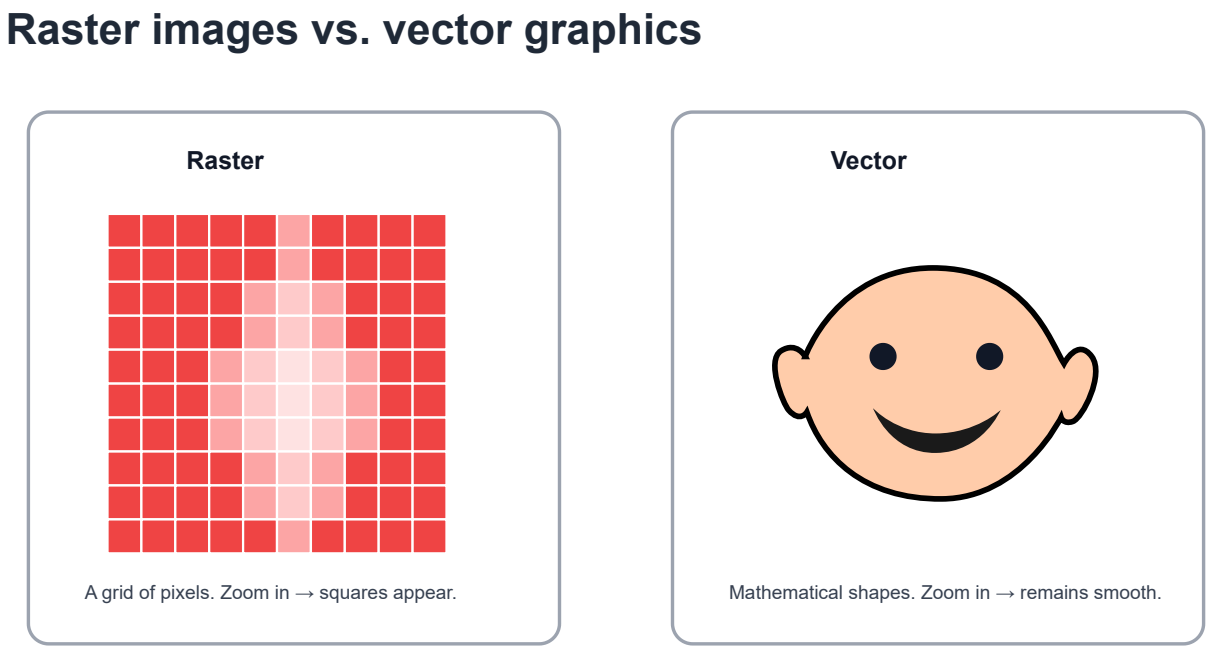
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3. Images: raster and vector

Most students are already familiar with image files such as JPG, PNG, GIF, and SVG. These files are not all represented in the same way.

Two fundamental representations are:

1. **Raster representation**
2. **Vector representation**



3.1 Raster graphics

A raster image is a rectangular grid of small picture elements called **pixels**.

Each pixel stores color information. For example, in an RGB image, a pixel may store three values:

```
Red   = 255
Green = 180
Blue  = 40
```

Together these values produce a particular color.

Raster graphics are common in:

- digital photographs,
- screenshots,
- scanned documents,
- video frames,
- texture images in games,
- medical images,
- satellite images.

The major advantage of raster graphics is that they can represent complex real-world images. The major limitation is that zooming too much reveals the pixel grid.

3.2 Vector graphics

A vector image describes shapes mathematically using geometric primitives such as:

- points,
- lines,
- curves,
- rectangles,
- circles,
- paths,
- polygons.

Vector graphics are common in:

- logos,
- icons,

- technical drawings,
- fonts,
- maps,
- diagrams,
- SVG illustrations.

The major advantage of vector graphics is scalability. If a logo is represented by curves and shapes, it can be enlarged without becoming blocky.

Trivia: why fonts look smooth

Modern fonts are usually stored using vector outlines, not as fixed bitmap images. When the same letter is displayed at different sizes, the computer converts the mathematical outline of the letter into pixels at the required size. This conversion process is called **rasterization**.

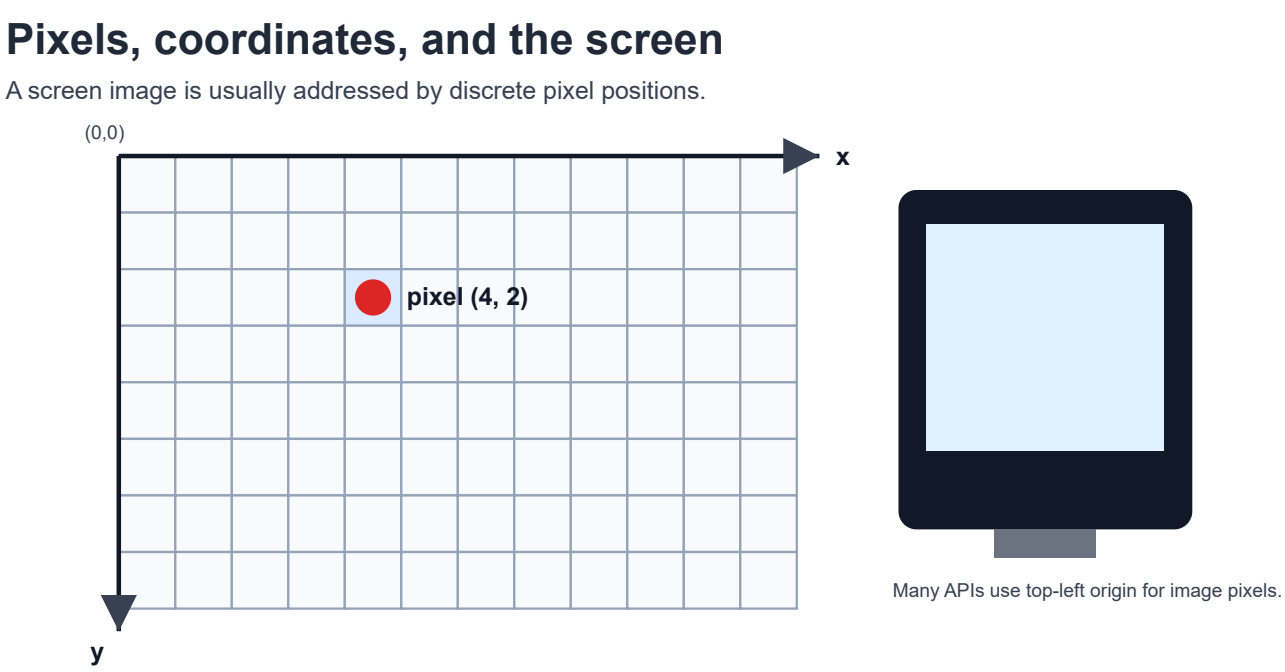
That is why the same font can appear smooth in a small document, on a large poster, and on a high-resolution phone screen.

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4. Pixels and coordinates

A display screen can be imagined as a grid of pixels. Each pixel has a position.

In many image-processing and graphics systems, the origin `(0, 0)` is placed at the **top-left corner** of the image. The x-coordinate increases to the right, and the y-coordinate increases downward.



This may feel different from the Cartesian coordinate system used in mathematics, where the y-axis often increases upward. Computer graphics frequently moves between different coordinate systems, so students should be comfortable with coordinate conversion.

Important idea

A mathematical line can have infinitely many points. A screen line cannot. A screen line must be approximated by a finite set of pixels.

This is why we need **scan conversion algorithms** such as:

- point scan conversion,
- line scan conversion,
- circle scan conversion,
- ellipse scan conversion,
- polygon filling.

These topics will become important in later lectures.

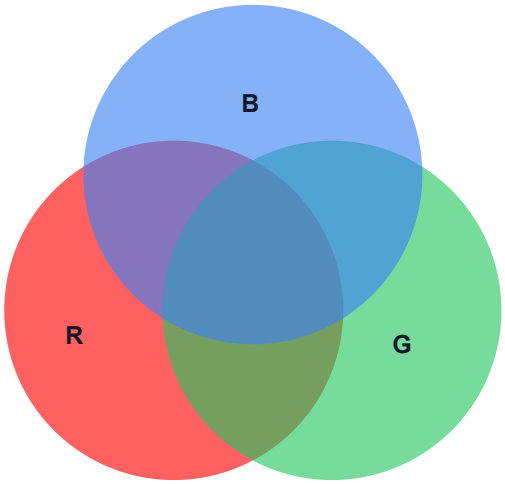
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5. Color as numbers

Color on a digital display is usually represented numerically. A common model is the **RGB color model**, where colors are produced by mixing red, green, and blue light.

RGB: color as numbers

Most screens mix red, green, and blue light.



One pixel

R = 255

G = 180

B = 40

Numbers become visible light.

For an 8-bit-per-channel RGB image:

- Red can range from 0 to 255,
- Green can range from 0 to 255,
- Blue can range from 0 to 255.

So one pixel can represent:

```
256 × 256 × 256 = 16,777,216 possible colors
```

This is often called **24-bit color** or **true color**.

Trivia: screen color is additive

Paint mixing and screen color mixing are different.

- Paint uses a mostly **subtractive** process: pigments absorb some wavelengths and reflect others.
- Screens use an **additive** process: red, green, and blue light are added together.

On a screen:

```
Red + Green = Yellow
Red + Blue = Magenta
Green + Blue = Cyan
```

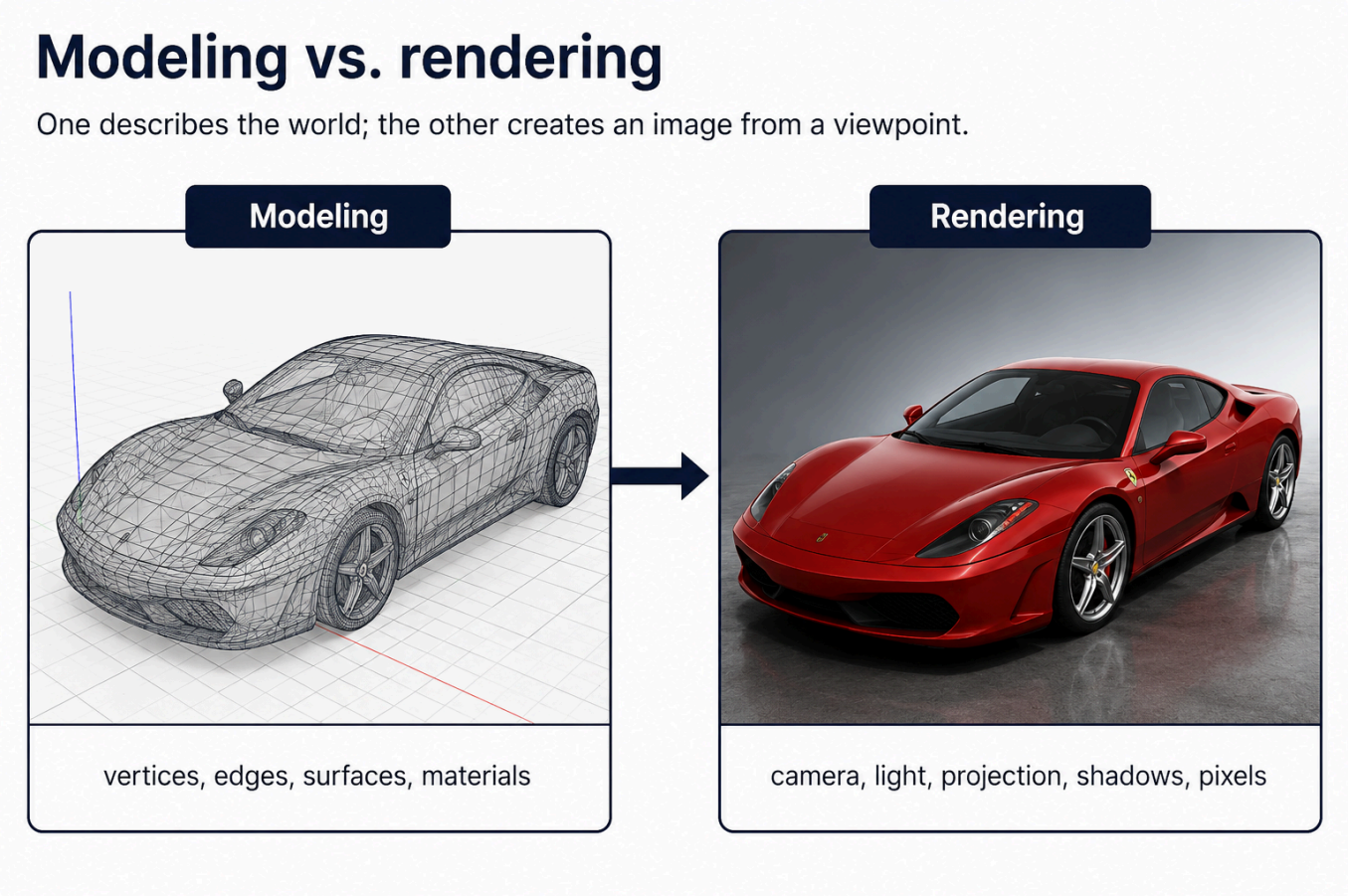
```
Red + Green + Blue = White
No light = Black
```

This is why the RGB model is natural for displays.

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6. Modeling and rendering

Two important words in computer graphics are **modeling** and **rendering**.



6.1 Modeling

Modeling means creating a description of the objects or scene.

A model may include:

- object shape,
- object size,
- position,
- orientation,
- material,
- surface color,
- texture,

- animation information.

For example, a 3D model of a chair may contain vertices, edges, faces, surface normals, texture coordinates, and material properties.

6.2 Rendering

Rendering means generating an image from a model.

Rendering considers:

- camera position,
- projection type,
- light sources,
- material response,
- visibility,
- shadows,
- reflections,
- final pixel colors.

In short:

```
Modeling describes the world.
Rendering creates an image of the world.
```

Example

A game developer may create a 3D model of a car once. During gameplay, the rendering system draws that car many times per second from different camera angles.

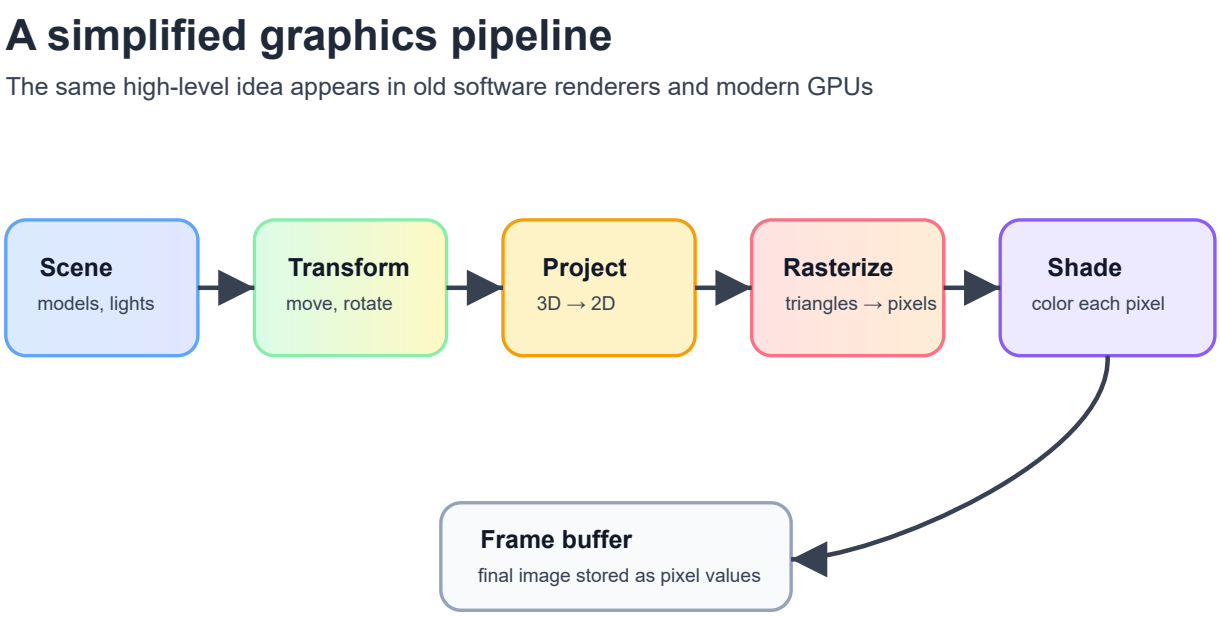
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7. The graphics pipeline

A graphics pipeline is a sequence of steps that converts scene descriptions into final pixels.

A simplified version is:

```
Scene data → Transformation → Projection → Rasterization → Shading → Frame buffer → Display
```



7.1 Scene data

The scene contains objects, lights, cameras, and material information.

7.2 Transformation

Objects are moved, rotated, scaled, or positioned in the scene. Transformations are usually represented using matrices.

7.3 Projection

A 3D scene is converted into a 2D view. Projection is the mathematical reason a 3D cube can appear on a flat monitor.

7.4 Rasterization

Geometric shapes such as lines and triangles are converted into pixels.

For example:

```
A triangle in mathematical space → a set of screen pixels covered by that triangle
```

7.5 Shading

The color of each pixel is computed. Shading may consider material color, light direction, texture maps, surface normals, and shadows.

7.6 Frame buffer

The generated pixel values are stored in a memory area called the frame buffer. The display reads from this buffer to show the final image.

Trivia: why games need high frame rates

If a game runs at 60 frames per second, the computer must generate a new image roughly every:

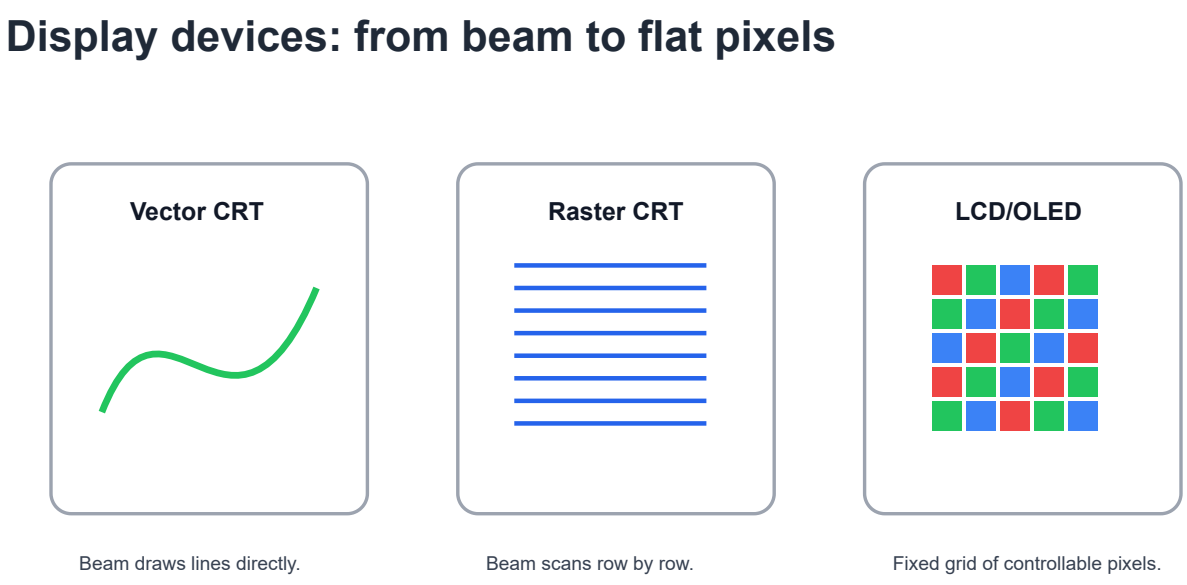
```
1 / 60 second ≈ 16.67 milliseconds
```

That includes input handling, physics, animation, AI, geometry processing, shading, and display preparation. Real-time graphics is therefore a race against time.

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8. Display devices: a brief view

Computer graphics has always been strongly connected with display technology.



8.1 Vector displays

Early graphics displays could draw lines directly using an electron beam. They were good for wireframe drawings but not ideal for filled images.

8.2 Raster displays

Raster displays scan the screen row by row. This idea became dominant because it is suitable for images, filled polygons, text, and video.

8.3 Modern flat-panel displays

Modern LCD and OLED displays use a fixed grid of pixels. Each pixel or subpixel can be controlled electronically.

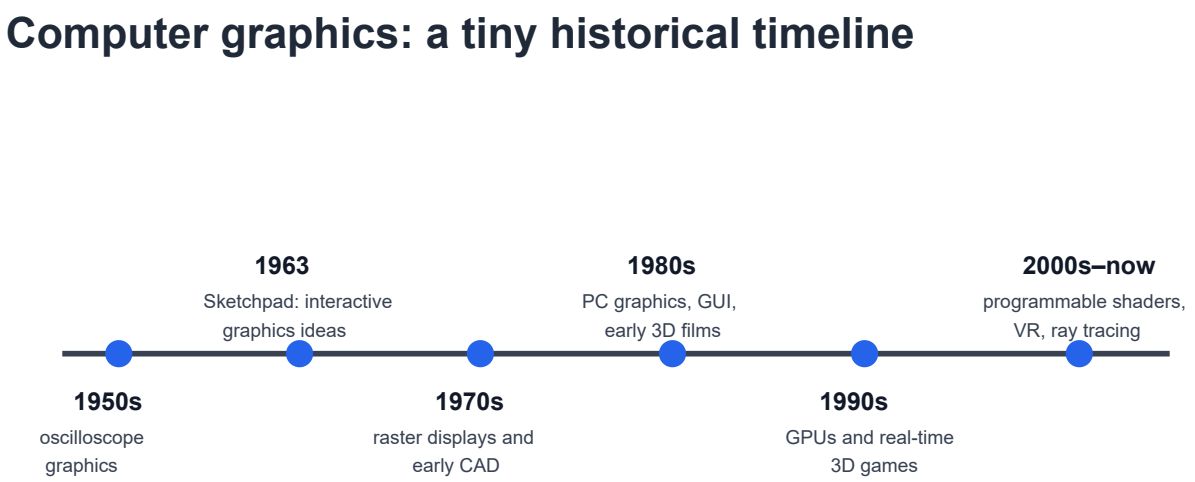
Trivia: the word “pixel”

The word **pixel** comes from **picture element**. A pixel is not always a tiny square physically, but in most introductory graphics discussions it is convenient to imagine it as one cell in a rectangular grid.

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9. A very short history of computer graphics

Computer graphics did not suddenly appear with modern games. Its roots go back to early display systems, scientific visualization, CAD, simulation, and human-computer interaction.



Some historical milestones:

1950s: oscilloscope and vector graphics

Early systems used displays similar to oscilloscopes. Graphics were often line-based rather than pixel-grid based.

1960s: interactive graphics

Ivan Sutherland's **Sketchpad** is often remembered as a landmark system because it demonstrated interactive drawing, constraints, and graphical manipulation at a time when computers were mostly text-oriented.

1970s: raster displays and CAD growth

Raster displays became more practical. Computer-aided design began to grow in engineering and manufacturing.

1980s: personal computers and graphical interfaces

Graphical user interfaces became increasingly important. Computer graphics moved closer to everyday users.

1990s: real-time 3D and GPUs

Dedicated graphics hardware made real-time 3D graphics practical for games and visualization.

2000s to now: programmable graphics, VR, and ray tracing

Modern GPUs are programmable parallel processors. They are used not only for graphics but also for machine learning, scientific computing, and simulation.

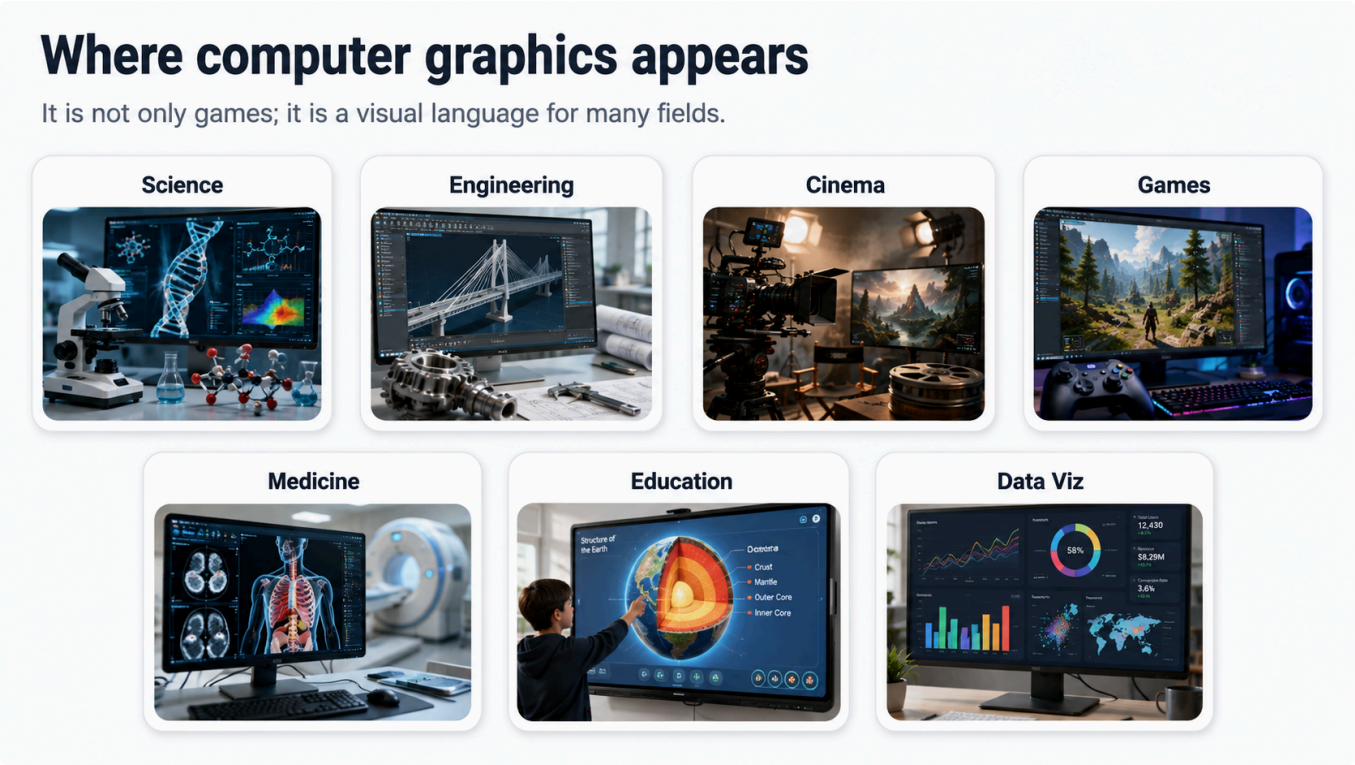
Trivia: graphics helped create modern interaction

Many ideas now considered normal—windows, icons, menus, pointers, direct manipulation—were historically tied to graphics research. Computer graphics is therefore closely related to human-computer interaction.

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10. Application areas of computer graphics

Computer graphics is used almost everywhere visual communication is needed.

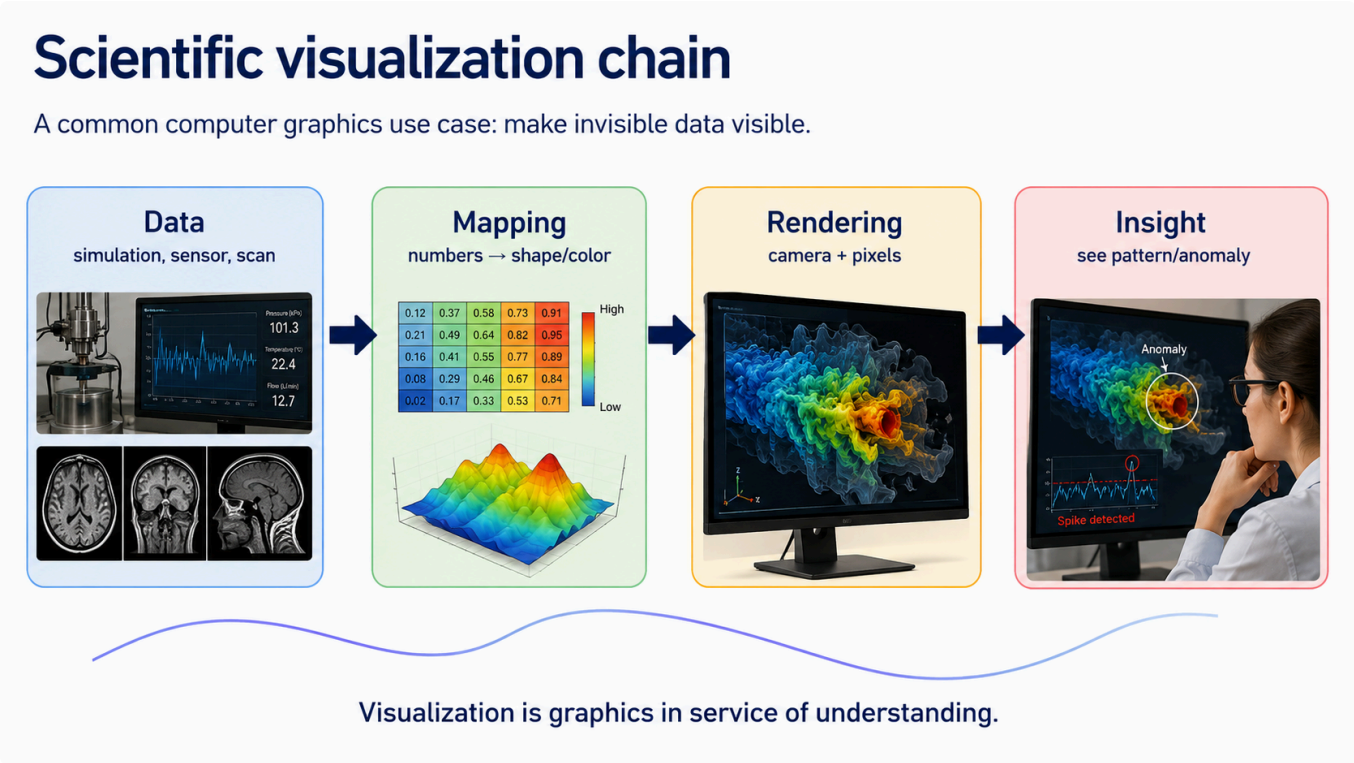


10.1 Scientific visualization

Scientists often work with data that cannot be understood easily as raw numbers. Graphics can reveal patterns.

Examples:

- weather simulation,
- fluid flow visualization,
- molecular structures,
- astronomical data,
- oceanographic data,
- climate models.



In scientific visualization, the goal is not always photorealism. The goal is often **insight**.

10.2 Engineering and CAD

Computer-aided design uses graphics to design and inspect objects before manufacturing.

Examples:

- buildings,
- bridges,
- machines,
- circuit boards,
- ships,
- aircraft parts.

Graphics allows engineers to view objects from different angles, measure geometry, simulate motion, and detect design issues.

10.3 Medicine

Medical graphics helps doctors and researchers visualize the human body.

Examples:

- CT scan visualization,
- MRI volume rendering,

- surgical planning,
- dental imaging,
- anatomy education.

A medical scan is not directly a photograph. It is often a set of measured values reconstructed and displayed visually.

10.4 Education

Graphics can make abstract concepts visible.

Examples:

- geometry animations,
- physics simulations,
- interactive maps,
- molecular models,
- algorithm visualizations.

For computer science education, visualizing sorting, graph traversal, memory layout, or recursion can help students understand processes that are otherwise invisible.

10.5 Entertainment and games

Games, animation, visual effects, and virtual production rely heavily on computer graphics.

Important areas include:

- character modeling,
- environment design,
- lighting,
- animation,
- particle effects,
- real-time rendering,
- physics-based motion,
- camera control.

10.6 Data visualization

Data visualization uses graphics to communicate data.

Examples:

- bar charts,

- line charts,
- heat maps,
- network diagrams,
- dashboards,
- geographic information systems.

A good visualization can make a trend visible immediately. A poor visualization can mislead the audience.

10.7 Graphical user interfaces

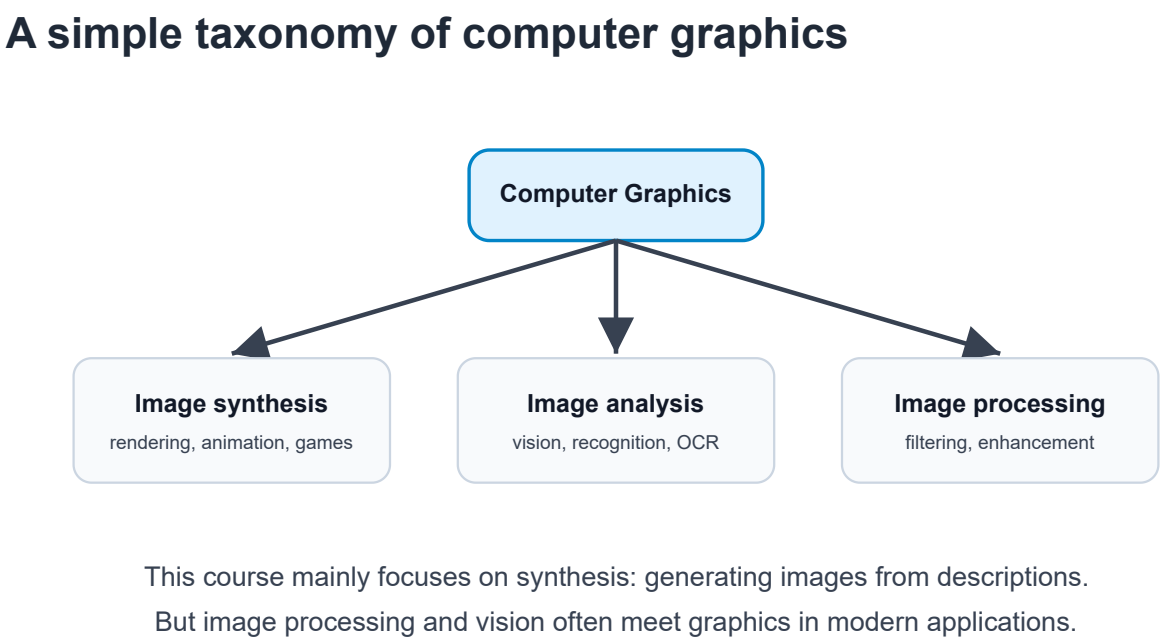
Every button, icon, window, scrollbar, cursor, and menu is a computer graphics object at some level.

Even ordinary software depends on graphics to communicate with users.

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11. Relationship with other fields

Computer graphics is often placed near other visual computing fields.



This course mainly focuses on synthesis: generating images from descriptions.
But image processing and vision often meet graphics in modern applications.

11.1 Computer graphics vs. image processing

Computer graphics usually creates images from models or descriptions.

Image processing usually starts with an existing image and modifies or analyzes it.

Example:

```
Computer graphics: create an image of a 3D chair.
Image processing: sharpen a photograph of a chair.
```

11.2 Computer graphics vs. computer vision

Computer graphics asks:

```
Given a scene, how do we create an image?
```

Computer vision asks:

```
Given an image, what scene or object does it represent?
```

In a simple sense, graphics and vision are opposite directions:

```
Graphics: scene → image
Vision: image → scene understanding
```

11.3 Computer graphics and multimedia

Multimedia systems combine text, audio, image, animation, video, and interaction. Computer graphics provides many of the visual components used in multimedia applications.

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12. The course roadmap

This course will move from basic representation to more advanced 2D and 3D graphics.

The major topics are:

1. **Introduction to computer graphics**
2. **Image representation**
3. **Scan conversion**
4. **Two-dimensional transformation**
5. **Two-dimensional viewing and clipping**
6. **Three-dimensional transformation**
7. **Projection**
8. **Three-dimensional viewing and clipping**
9. **Geometric representation**

10. **Hidden surface removal**
11. **Computer animation**

A useful way to think about the course is:

```
Pixels → Shapes → Transformations → Viewing → 3D → Visibility → Animation
```

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13. Key terms

Term	Meaning
Pixel	Smallest addressable element of a raster image or display grid
Raster image	Image represented as a grid of pixels
Vector graphics	Image represented using mathematical shapes
Frame buffer	Memory area that stores pixel values for display
Modeling	Creating a mathematical/computational description of objects
Rendering	Generating an image from a scene description
Rasterization	Converting geometric primitives into pixels
Projection	Mapping 3D objects onto a 2D view plane
Transformation	Changing position, size, or orientation of objects
Visualization	Using graphics to make data understandable

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14. Short worked examples

Example 1: number of pixels in a Full HD image

A Full HD image has resolution:

```
1920 × 1080
```

Total pixels:

```
1920 × 1080 = 2,073,600 pixels
```

So a single Full HD frame contains more than two million pixels.

If each pixel uses 24 bits = 3 bytes, the uncompressed size is:

```
2,073,600 × 3 = 6,220,800 bytes ≈ 5.93 MiB
```

A video at 60 frames per second would require a huge amount of raw data if not compressed.

Example 2: why a diagonal line looks stair-stepped

A perfect mathematical diagonal line is continuous. A display is discrete. So when the line is drawn on a grid, the algorithm must choose nearby pixels.

This creates a stair-step effect called **aliasing**.

Later we will study **anti-aliasing**, which reduces this visual jaggedness by using intermediate colors or intensities.

Example 3: 3D to 2D illusion

When a cube is drawn on paper or on screen, it is not physically three-dimensional. The display is flat. But by drawing projected edges and using shading, the brain interprets depth.

Computer graphics uses this perceptual ability to create 3D visual experiences on 2D screens.

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15. In-class activities

Activity 1: classify the object

For each item, decide whether it is mainly raster, vector, or both.

Item	Likely representation
A phone camera photo	Raster
A university logo in SVG format	Vector
A scanned handwritten note	Raster
A Google Maps road line	Vector-like data rendered to raster screen
A game texture	Raster
A font letter before display	Vector outline
A font letter after display	Raster pixels

Activity 2: find graphics around you

Ask students to list five examples of computer graphics visible in the classroom or on their phones.

Possible answers:

- app icons,
- wallpaper,
- browser rendering,
- video player interface,
- digital clock display,
- presentation slides,
- maps,
- emojis,
- camera preview,
- QR code display.

Activity 3: pipeline thinking

Take a simple scene: a red ball on a table.

Ask students:

1. What data is needed to model the scene?
2. Where is the camera?
3. Where is the light?
4. Which parts of the ball are visible?
5. Which pixels should be red?
6. Where should the shadow appear?

This helps students see that even a simple image involves many decisions.

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16. Common misconceptions

Misconception 1: computer graphics means only animation

Animation is one application, but computer graphics includes static images, diagrams, interfaces, visualization, CAD, games, simulation, and more.

Misconception 2: 3D graphics means the screen is 3D

The screen is usually 2D. The scene may be represented in 3D, but the final output is projected into a 2D image.

Misconception 3: higher resolution always means better image quality

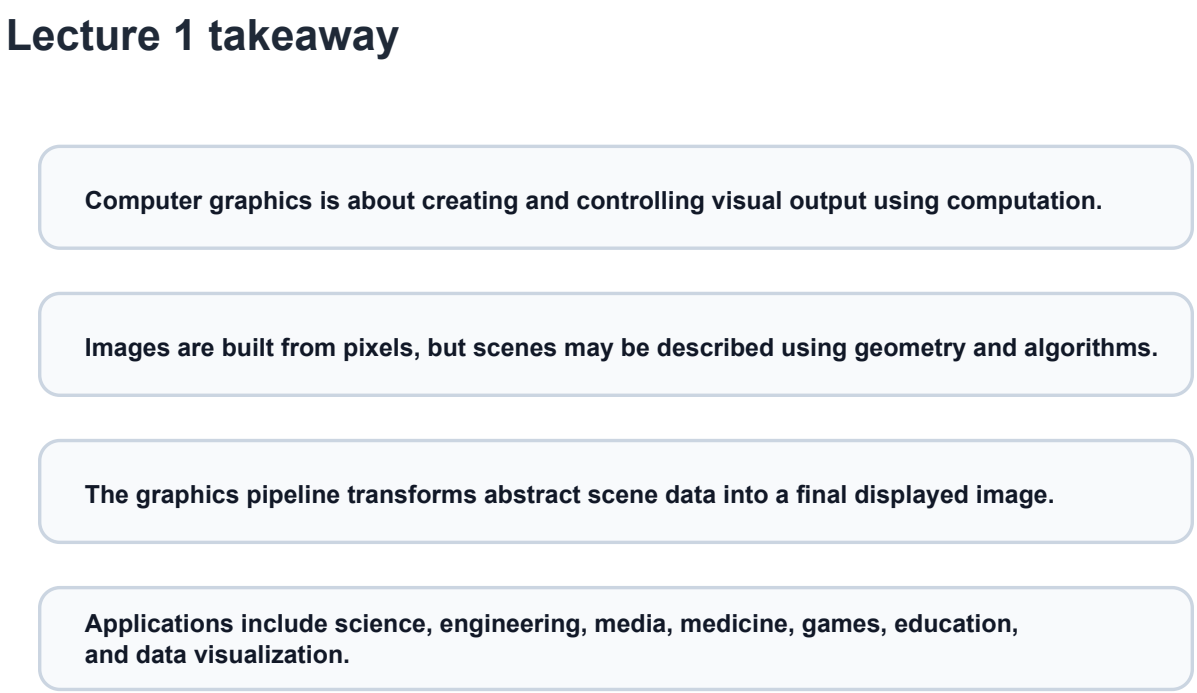
Resolution matters, but quality also depends on color accuracy, contrast, compression, lighting, sampling, display technology, and viewing distance.

Misconception 4: realistic graphics is always the goal

Many graphics applications prioritize clarity over realism. A metro map, medical visualization, or engineering diagram may be intentionally simplified.

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17. Lecture summary



Computer graphics is the computational production and control of visual output. It connects geometry, algorithms, hardware, color, perception, and design. The course begins with pixels and image representation, then moves toward scan conversion, transformations, viewing, projection, hidden surfaces, and animation.

The central journey of the course is:

```
Abstract description → mathematical processing → visible image
```

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18. Review questions

1. Define computer graphics in your own words.
2. What is the difference between raster and vector graphics?

3. Why does a raster image become blocky when enlarged too much?
4. What is a pixel?
5. What is the RGB color model?
6. What is the difference between modeling and rendering?
7. What is the purpose of the frame buffer?
8. Why is projection necessary in 3D computer graphics?
9. Name five application areas of computer graphics.
10. Explain the difference between computer graphics and image processing.

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19. Short exam-style questions

Question 1

Explain why computer graphics is considered an interdisciplinary field.

Expected answer outline: It uses mathematics, algorithms, data structures, hardware, physics, design, and human perception to generate and control images.

Question 2

Differentiate between raster graphics and vector graphics with examples.

Expected answer outline: Raster uses pixels and is suitable for photos; vector uses mathematical primitives and is suitable for logos, diagrams, and fonts.

Question 3

Describe the basic stages of a simplified graphics pipeline.

Expected answer outline: Scene data, transformation, projection, rasterization, shading, frame buffer, display.

Question 4

Why is a line-drawing algorithm necessary in raster graphics?

Expected answer outline: A mathematical line is continuous, but a raster display is discrete; an algorithm chooses which pixels best approximate the line.

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20. Suggested reading

1. Steven Harrington, *Computer Graphics: A Programming Approach*.
2. Donald Hearn and M. Pauline Baker, *Computer Graphics*.

3. *Schaum's Outline of Computer Graphics*.
4. Any introductory OpenGL/WebGL tutorial may be used later for practical understanding, but the mathematical concepts should be understood independently of a specific library.

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21. Instructor notes

- Start the class by showing familiar examples: phone UI, game screenshot, Google Maps, medical scan, movie VFX, engineering CAD.
- Ask students whether these are graphics, image processing, multimedia, or visualization.
- Emphasize that this course is algorithmic, not only tool-based.
- Students should understand that later topics such as scan conversion and clipping are foundations of how graphics systems work internally.
- For engagement, use a quick demo: zoom into a raster image and compare it with an SVG icon.